Times New Roman

Bayesian Neural Surrogates Unify Uncertainty Propagation, Sensitivity Attribution and Calibration for Coupled Multi-Physics Reactor Analysis

Yinuo Chen[1,2], Sicheng Wang[2,3], Xiaojing Liu[1,2*], Tengfei Zhang[1,2*]

1. School of Nuclear Science and Engineering, Shanghai Jiao Tong University, Shanghai 200240, China
2. Shanghai Digital Nuclear Reactor Technology Integration Innovation Center, Shanghai 200240, China
3. College of Smart Energy, Shanghai Jiao Tong University, Shanghai 200240, China

* Corresponding authors: zhangtengfei@sjtu.edu.cn; xiaojingliu@sjtu.edu.cn

NC Draft · April 20, 2026

Abstract

We train a Bayesian neural network (BNN) surrogate on coupled OpenMC–FEniCS data of a heat-pipe-cooled microreactor and use its posterior predictive distribution as the single probabilistic object on which forward uncertainty propagation, Sobol sensitivity attribution and MCMC posterior calibration all operate, without retraining or recalibration between analyses. Two-way neutronic–thermal–structural coupling damps stress variability by approximately 30% relative to a decoupled prediction; stress and reactivity are governed by largely separate material parameters ($E_{\text{intercept}}$ and α_{base}, respectively); and across 18 benchmark calibrations the posterior contracts toward observation-compatible regions with 90%-CI coverage of 0.861. Each posterior predictive draw costs approximately 1.43×10^5 times less than a coupled high-fidelity solve, enabling the full uncertainty workflow to run end-to-end on a single workstation. The posterior-predictive framework is directly applicable to other coupled multi-physics systems.

1 Introduction

Microreactors—compact, transportable fission systems rated at 1–20 $\mathrm{MW_e}$—are being developed for decentralized power generation in remote communities, defense installations and space missions[1,11]. NASA and DOE are advancing fission surface-power systems for sustained lunar and planetary operations[12], while rapidly growing electricity demand from AI-intensive data centres reinforces the case for distributed, dispatchable zero-carbon generation[13]. Unlike conventional large-scale plants, these designs concentrate fission power, heat removal and structural loading within a compact monolithic core where neutronics, heat transfer and structural mechanics are tightly coupled through two-way feedback[1,2]. A single uncertain material parameter can simul-

taneously alter the neutronic balance, the thermal field and the stress state; the closed feedback loop may amplify or damp the resulting variability in ways that single-physics analyses cannot capture. The uncertainty problem in these systems is therefore inherently coupled.

High-fidelity coupled solvers now exist for these systems: MOOSE-based toolchains handle neutronics–thermo-mechanics–heat-pipe coupling for KRUSTY-class and larger designs[2,3], and thermal–structural analyses have established stress concentration as a persistent safety concern in monolithic-core configurations[4,5]. Surrogate-assisted probabilistic modules in the MOOSE ecosystem[6,7] and digital-twin pipelines[8] address forward propagation, sensitivity or calibration individually, but each uses a separately constructed model with no shared probabilistic basis between analyses.

Running the full probabilistic workflow—forward uncertainty propagation, global sensitivity analysis and Bayesian calibration—directly on the coupled solver is impractical at $\sim 10^3$ s per solve. Gaussian-process and polynomial-chaos surrogates face dimensionality constraints in nonlinear coupled settings; physics-informed neural networks require a closed residual form that iterative solver exchange does not supply; deterministic neural networks produce only point predictions. None of these provides the coherent predictive distribution that all three analyses require.

A Bayesian neural network (BNN) fills this gap. By placing a distribution over the network weights, a BNN yields a posterior predictive distribution that can be sampled at negligible cost once trained. This single distribution serves as the shared probabilistic layer on which forward propagation, variance-based sensitivity attribution and observation-conditioned posterior calibration all operate, with no retraining or recalibration between them.

This paper presents a BNN-based posterior-predictive framework for end-to-end probabilistic analysis of coupled multi-physics systems. The framework unifies forward uncertainty propagation, Sobol sensitivity decomposition and MCMC posterior calibration on a single predictive layer trained on coupled OpenMC–FEniCS data of a heat-pipe-cooled reactor. It reveals that two-way coupling damps stress variability by approximately 30%, that stress and reactivity are governed by largely separate parameter pathways, and that the posterior calibration contracts the material-parameter distribution toward observation-compatible regions—all at five orders of magnitude less cost per evaluation than a coupled high-fidelity solve.

2 Results

2.1 Surrogate accuracy and calibration

The framework rests on a single BNN that predicts all 15 coupled and decoupled outputs simultaneously; if this surrogate is inaccurate, the downstream propagation, sensitivity and calibration results are unreliable. We evaluated the physics-regularized BNN alongside a reference (unconstrained) BNN and two conventional baselines—MC-Dropout and a five-member deep ensemble—on 501 held-out test samples (Table 1, Fig. 1). Results for the physics-regularized BNN are reported as mean ± s.d. across 5 random seeds; ablation across both BNN variants is in Supplementary Table S3.

On coupled peak global stress the physics-regularized BNN achieves $R^2 = 0.935 \pm 0.007$ (RMSE 7.63 ± 0.47 MPa), comparable to MC-Dropout ($R^2 = 0.934$) and deep ensemble ($R^2 = 0.934$).

The BNN advantage is most pronounced on temperature outputs: for maximum fuel temperature the BNN RMSE is 4.26 ± 0.18 K versus 4.50 K for both baselines, and the expected calibration error (ECE) is 0.027 versus 0.183 (MC-Dropout) and 0.190 (deep ensemble)—a five- to six-fold improvement in calibration quality (Fig. 1A, C).

The 90% predictive intervals are conservatively calibrated across all outputs (PICP 0.937–1.000), and incorporating physics-prior monotonicity constraints tightens the intervals relative to the unconstrained reference without reducing coverage (Supplementary Table S3). The continuous ranked probability score (CRPS) is lowest for the physics-regularized BNN on all five primary outputs (Table 1).

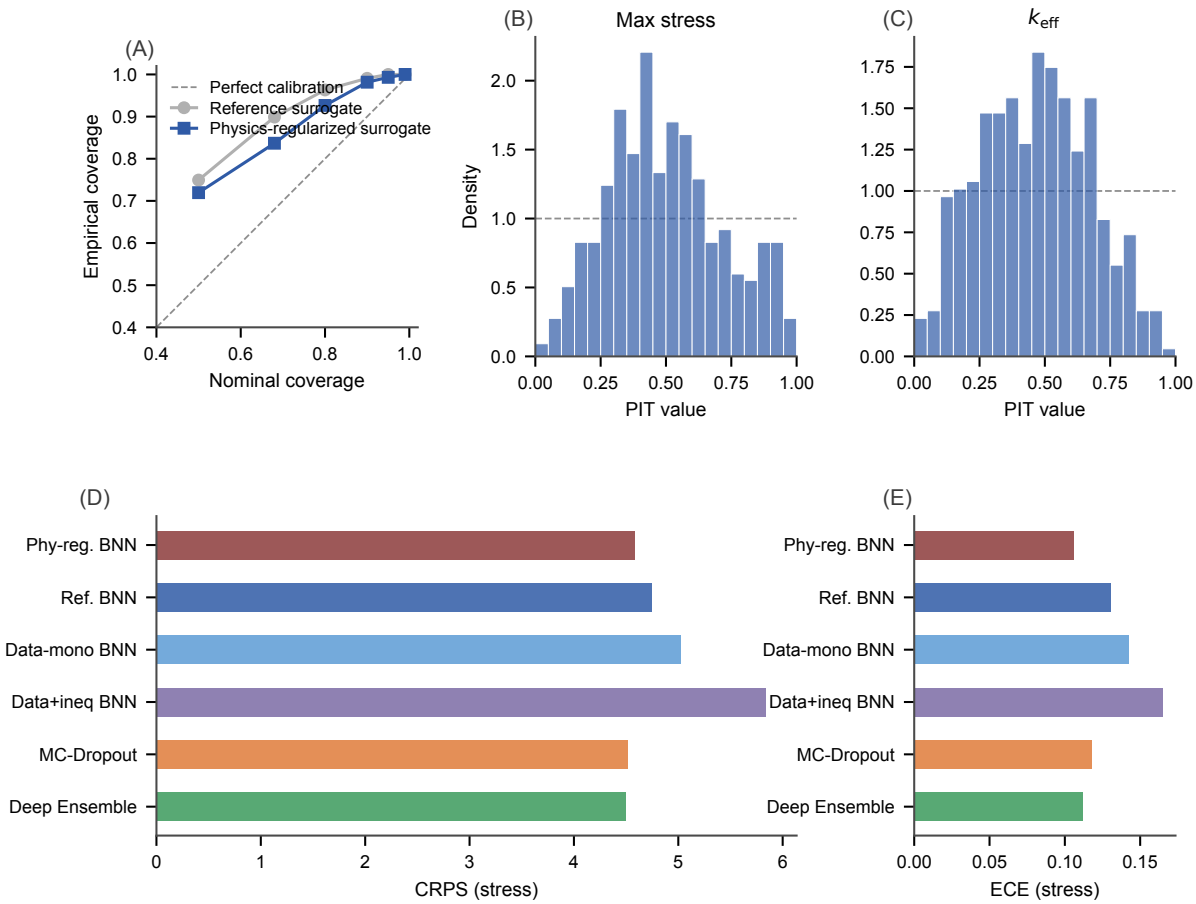

Figure 1: **Surrogate accuracy and uncertainty calibration.** (**A**) Predicted vs. true values for five primary coupled outputs (physics-regularized BNN, 501 test samples). (**B**) Residual distributions. (**C**) Calibration curves comparing BNN, MC-Dropout and deep ensemble. The BNN achieves five- to six-fold lower ECE than the baselines on temperature outputs.

2.2 Physics consistency

We verified the BNN's adherence to known physical relationships through two tests: monotonicity verification against 15 input–output pairs derived from the constitutive laws, and ordering/non-negativity constraint satisfaction (Fig. 2A). For all high-confidence monotonicity pairs—including the dominant stress drivers $E_{\mathrm{intercept}}$ (+), α_{base} (+), and k_{ref} (−)—both BNN variants achieved zero violation rates on the 501 held-out test samples using discrete $\pm\delta$ perturbation testing. The

Table 1: **Surrogate accuracy on five primary coupled outputs.** Physics-regularized BNN: mean ± s.d. across 5 seeds; external baselines: single run. Bold CRPS indicates best per output.

Output	R^2	RMSE	PICP	ECE	CRPS
Physics-regularized BNN					
k_{eff}	0.633 ± 0.258	0.0006 ± 0.0004	0.981 ± 0.007	0.123	**0.0002**
Max. fuel temp.	0.621 ± 0.044	4.26 ± 0.18 K	0.936 ± 0.015	0.027	**2.39**
Max. monolith temp.	0.622 ± 0.044	4.80 ± 0.20 K	0.942 ± 0.013	0.031	**2.70**
Max. global stress	0.935 ± 0.007	7.63 ± 0.47 MPa	0.987 ± 0.007	0.106	**4.58**
Wall expansion	0.993 ± 0.001	0.0020 ± 0.0002	1.000 ± 0.000	0.194	**0.0017**
MC-Dropout					
Max. fuel temp.	0.590	4.50 K	—	0.183	2.62
Max. global stress	0.934	7.66 MPa	—	0.118	4.52
Deep Ensemble (5 members)					
Max. fuel temp.	0.590	4.49 K	—	0.190	2.63
Max. global stress	0.934	7.64 MPa	—	0.112	4.50

only non-trivial violations occurred for the medium-confidence pair SS316_α → fuel temperature (36–54% across models), reflecting a weaker and potentially nonlinear relationship that the training data does not consistently support. All four models also satisfied ordering constraints (max ≥ average temperature) and non-negativity constraints (stress ≥ 0) with zero violations.

Decomposing total predictive variance into epistemic (weight-uncertainty) and aleatoric (observation-noise) components revealed that aleatoric uncertainty dominates across all outputs, accounting for 58–90% of total variance (Fig. 2B). The epistemic fraction was highest for wall expansion (∼42%) and stress (∼30%), indicating greater model-form uncertainty for these mechanically driven outputs. Physics regularization slightly reduced epistemic uncertainty for temperature predictions (epistemic fraction 10.1% vs 11.2–12.0% for the reference BNN), consistent with the expectation that monotonicity constraints restrict the space of plausible model structures.

2.3 Forward uncertainty propagation

Propagating 20,000 Latin hypercube samples through the BNN posterior predictive mean yields a coupled stress distribution with mean 161.7 MPa and standard deviation 31.9 MPa, versus a decoupled mean of 203.5 MPa and standard deviation of 45.6 MPa. Coupling reduces the stress standard deviation by approximately 30% and lowers the mean by approximately 42 MPa (Table 2, Fig. 3).

The coupled k_{eff} has mean 1.1036 and standard deviation approximately 77 pcm. The decoupled k_{eff} is effectively constant because, at the reference geometry, the sampled material parameters do not enter the neutronic cross-section tables; all coupled k_{eff} variability originates from thermal-expansion modification of core geometry.

2.4 Global sensitivity analysis

Sobol analysis (50 independent replications, $N_S = 4096$ per matrix; full index table in Supplementary Table S4) shows that stress and k_{eff} are governed by two largely separate physical pathways (Table 3, Fig. 4).

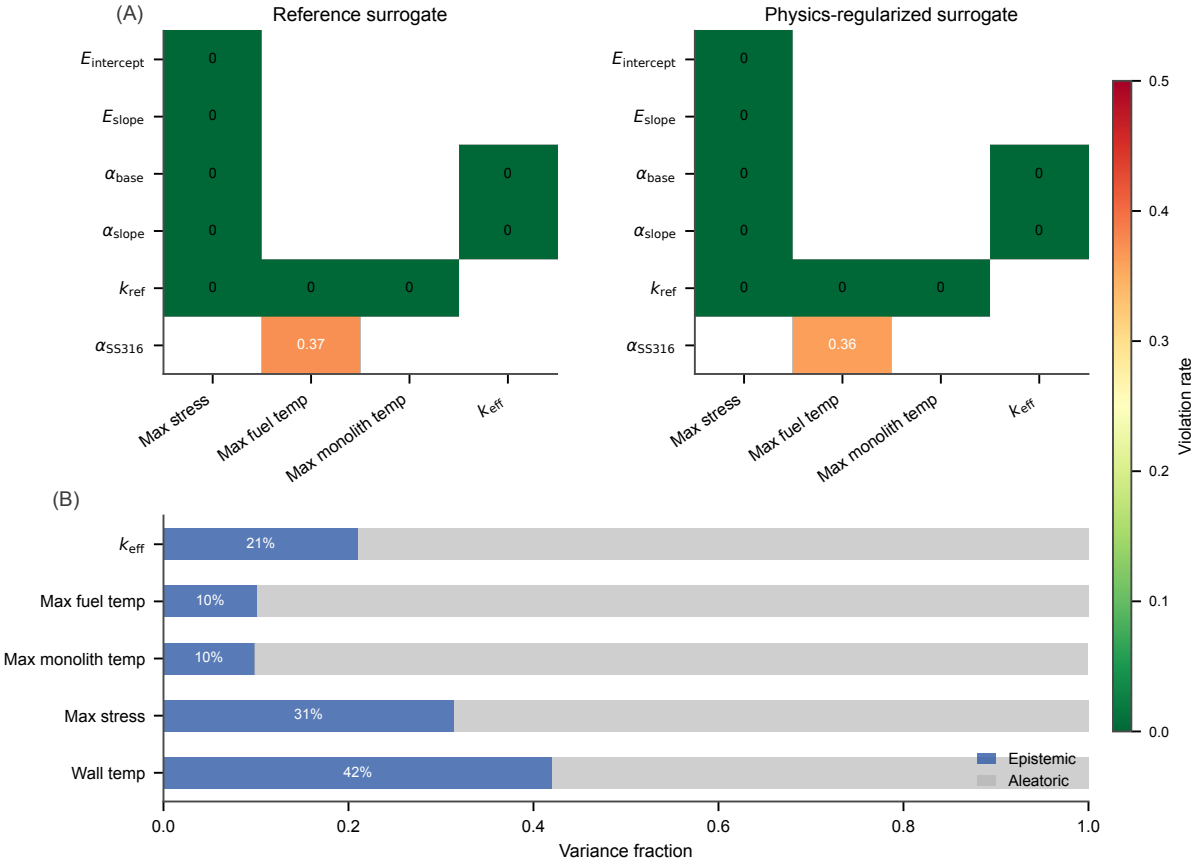

Figure 2: **Physics consistency verification.** (**A**) Monotonicity violation rates for 15 input–output pairs across both BNN variants; high-confidence pairs achieve zero violations. (**B**) Epistemic vs. aleatoric variance decomposition across five primary outputs.

Table 2: **Forward UQ summary: coupled vs. decoupled distributions** (20,000 LHS samples through BNN posterior predictive mean).

Output	Coupled mean	Coupled s.d.	Decoupled mean	Decoupled s.d.
Stress (MPa)	161.7	31.9	203.5	45.6
k_{eff}	1.1036	0.00077	∼constant	∼0

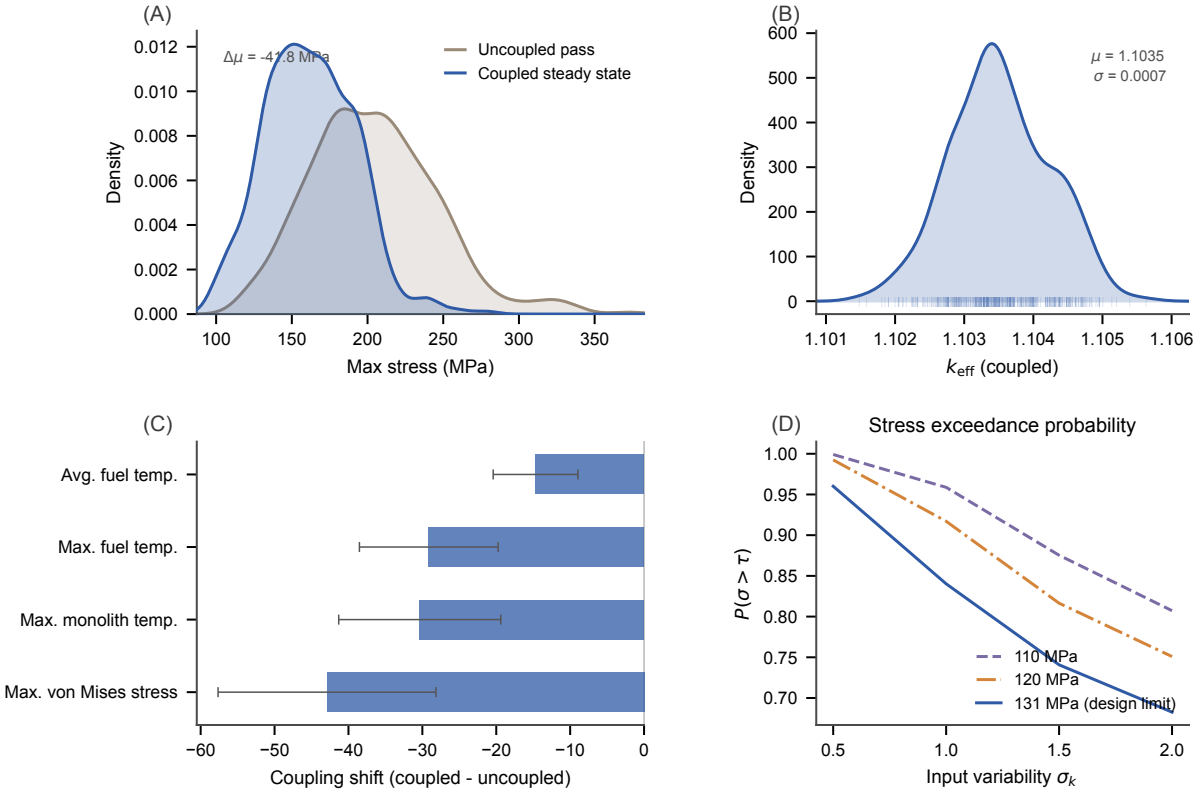

Figure 3: **Forward uncertainty propagation.** Coupled vs. decoupled output distributions for stress and k_{eff}. Two-way coupling reduces the stress standard deviation by ≈30% and introduces k_{eff} variability that is absent in the decoupled case.

Stress is rooted in the elastic-constitutive pathway. $E_{\text{intercept}}$ alone accounts for approximately 55% of the first-order variance ($S_1 = 0.548$, 90% CI [0.498, 0.572]), and the near-zero gap between its total-order and first-order indices indicates that it acts with minimal interaction. Secondary contributors—α_{base} ($S_1 = 0.210$), k_{ref} and ν—add a collective approximately 15%, while the remaining parameters are negligible. The dominance of $E_{\text{intercept}}$ is a direct consequence of the generalised Hooke's law: stiffness enters the stress field linearly.

k_{eff} is rooted in the thermal-expansion pathway. α_{base} accounts for approximately 77% of the first-order variance ($S_1 = 0.771$, 90% CI [0.747, 0.788]): it sets the magnitude of geometric expansion, which alters neutron leakage and shifts the neutronic balance. α_{slope} contributes a further approximately 19% ($S_1 = 0.195$), while no elasticity or conductivity parameter has a first-order effect on reactivity above the noise floor.

The two pathways share almost no dominant parameters. This separation means that stress and k_{eff} observations constrain different regions of the material-parameter space—an important constraint for any monitoring strategy that aims to reduce material-property uncertainty. Convergence analysis confirmed that these rankings stabilised by $N_S = 2048$, with standard deviations below 0.04 at the final sample size of 8192 (Supplementary Fig. S5). Both the reference and physics-regularized BNN yielded consistent rankings, indicating that the regularization does not distort the sensitivity structure.

Table 3: **First-order Sobol indices S_1 for stress and k_{eff}** (physics-regularized BNN, $N_S = 4096$, 50 replications, 90% CI). Only factors with CI lower bound > 0 are shown.

Parameter	Stress S_1 [90% CI]	k_{eff} S_1 [90% CI]
$E_{\text{intercept}}$	0.548 [0.498, 0.572]	—
α_{base}	0.210 [0.171, 0.254]	0.771 [0.747, 0.788]
ν	0.063 [0.034, 0.097]	—
k_{ref}	0.065 [0.014, 0.118]	—
α_{slope}	—	0.195 [0.112, 0.250]

2.5 Posterior calibration

Bayesian calibration uses the same BNN to update the prior parameter distribution given noisy observations. The four calibrated parameters—$E_{\text{intercept}}$, α_{base}, α_{slope} and k_{ref}—are the four that the Sobol analysis identified as covering the dominant variance channels for stress and k_{eff}. Metropolis–Hastings MCMC is run on 18 benchmark cases drawn from the test split (6 low / 6 near / 6 high stress; details in Methods) with 4 independent chains per case.

Chain convergence was assessed using rank-normalised split-$\hat{R}$ (Vehtari et al. 2021) and Geyer initial-positive-sequence effective sample size. Across all 72 parameter–case combinations for the physics-regularized BNN, $\hat{R}_{\max} = 1.017$ (well below the 1.1 threshold), $\text{ESS}_{\min} = 352$, and mean acceptance rate = 0.605 (Fig. 5A–B; Supplementary Table S5).

Across the 18 cases, the posterior 90%-CI coverage is 0.861—the observations successfully constrain the parameters that the Sobol analysis flagged as most influential. In higher-stress cases, the $E_{\text{intercept}}$ marginal shifts toward larger values, consistent with its dominant role in the elastic-

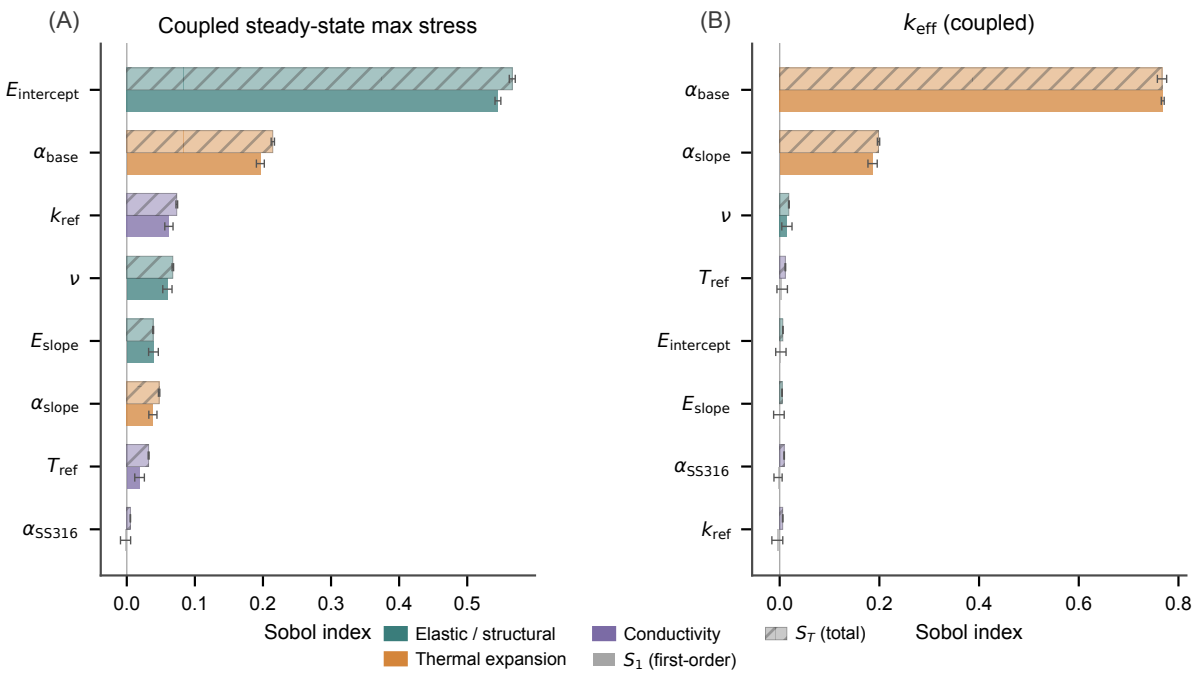

Figure 4: **Sobol sensitivity analysis.** First-order and total-order indices for stress and k_{eff} reveal two largely separate physical pathways: the elastic-constitutive pathway (dominated by $E_{\text{intercept}}$) for stress, and the thermal-expansion pathway (dominated by α_{base}) for k_{eff}.

constitutive pathway. The joint posterior of $E_{\text{intercept}}$ and α_{base} develops a compensation ridge: an increase in stiffness can be partially offset by a decrease in thermal expansion, leaving the stress product $\sigma \sim E \cdot \alpha \cdot \Delta T$ approximately invariant. This partial non-identifiability arises from the multiplicative structure and would not be visible in a single-output calibration (Fig. 5).

At the posterior-mean parameters of each case, the BNN-predicted stress tracks the true value with MAE = 10.4 MPa over a true-stress range of 102–198 MPa. Prior sensitivity analysis (6 prior configurations) and observation noise sensitivity analysis (noise fraction 0.5–10%) confirmed the robustness of posterior inference (Supplementary Figs. S7–S8, Tables S11–S12).

3 Discussion

3.1 Coupling feedback and uncertainty compression

The stress standard deviation drops from approximately 45.6 MPa in the decoupled response to approximately 31.9 MPa after coupling convergence, a roughly 30% reduction. The mechanism is the negative neutronic–thermal feedback characteristic of compact monolithic-core systems: thermal expansion modifies the core geometry, which shifts the neutronic balance and partially offsets the direct stress increase. What the BNN-based forward propagation establishes is the net effect of this feedback on the output distribution. A decoupled analysis does not see the feedback loop and therefore overestimates the parameter-driven stress variability.

The overestimate is conservative for safety-margin evaluation and cannot produce an unsafe conclusion, but it inflates the apparent penalty of material-property uncertainty. In practice, this

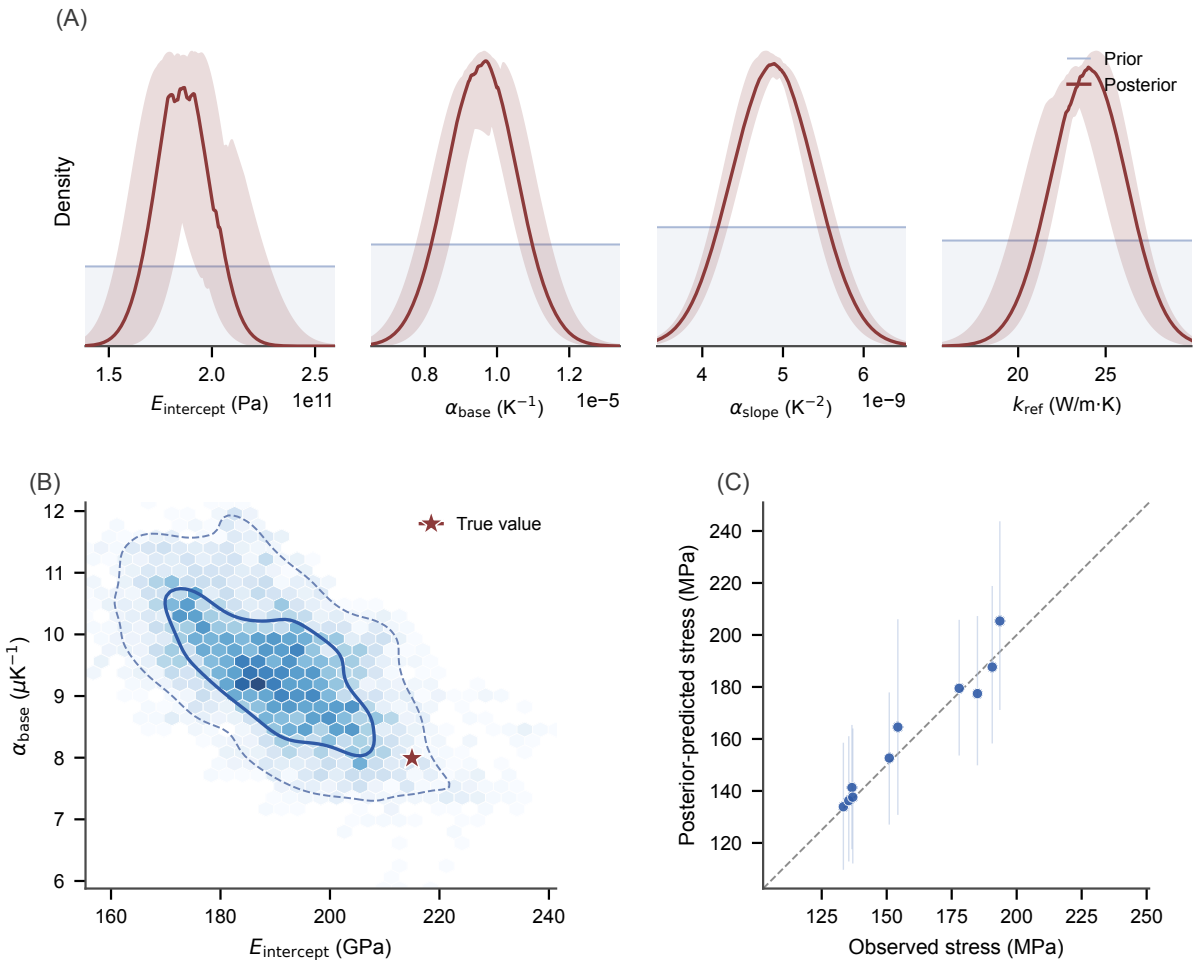

Figure 5: **Posterior calibration results.** (**A**) MCMC convergence diagnostics (split-$\hat{R}$ and ESS). (**B**) Acceptance rates across 18 benchmark cases. (**C**) Marginal posterior distributions for $E_{\text{intercept}}$ and α_{base} in representative low-, near-, and high-stress cases. (**D**) Joint posterior of $E_{\text{intercept}}$ vs. α_{base} showing the compensation ridge.

can translate into manufacturing tolerances stricter than the coupled response warrants.

3.2 BNN calibration and model selection

Both the unconstrained baseline and the physics-prior surrogate achieve zero monotonicity violations on the test set. Incorporating physics-prior monotonicity constraints modestly tightens predictive intervals without reducing coverage (stress CRPS 4.42 $\rightarrow$ 4.35; full ablation and small-sample data-efficiency analysis in Supplementary Notes G and H). The same posterior predictive distribution propagates consistently through forward UQ, Sobol attribution and MCMC calibration—a point-estimate surrogate would require separate uncertainty wrappers for each analysis with no guarantee of mutual consistency. Heteroscedastic noise modelling is retained over a homoscedastic alternative (Supplementary Note H).

Out-of-distribution analysis confirms that the BNN's epistemic uncertainty inflates appropriately in under-represented input regions: epistemic standard deviation increases by 7–21% in peripheral-tail subsets, while 90% coverage remains above 96% even in these regions (Supplementary Fig. S4). This Bayesian self-awareness is a structural advantage over frequentist baselines that lack a mechanism for signalling elevated model uncertainty.

3.3 Engineering implications

The channel separation identified by the Sobol decomposition (Section 2.4)—elastic-constitutive for stress, thermal-expansion for k_{eff}—translates into three practical consequences for the engineering workflow around this class of reactor.

Information-channel-aware material characterisation. Because the two output channels are governed by different material parameters, the characterisation campaign can be designed channel by channel. For stress-margin assessment, tightening the uncertainty on $E_{\text{intercept}}$—e.g. through coupon tensile tests at operating temperature—is the single highest-leverage action; for reactivity stability, dilatometry to refine α_{base} is the priority. Because these two measurements target different experimental setups, they can proceed in parallel without competing for the same facility.

Iterative updating. When measured or monitored data become available—whether from prototype testing, post-irradiation examination or in-service monitoring—the posterior calibration folds that information into the same surrogate without retraining. The updated parameter distribution propagates immediately to all 15 outputs, producing progressively tighter safety-margin estimates as characterisation accumulates. If a posterior update reveals that a safety margin is tighter than initially expected, the Sobol ranking identifies which parameter to prioritise in the next round of characterisation.

Multi-output experimental design. The near-zero parameter overlap between the stress and k_{eff} pathways means that a monitoring strategy based on a single composite metric (e.g. a weighted safety index) would lose information. Stress observations constrain elastic properties; k_{eff} observations constrain thermal-expansion properties. A calibration protocol that combines

both provides stronger parameter identification than either alone, and the Sobol decomposition quantifies how much each observation type contributes.

These indications are model-driven and require verification on the high-fidelity solver before adoption as design directives (see Section 3.5).

3.4 Computational efficiency

The BNN produces one posterior-predictive draw, averaged over 50 Monte Carlo weight samples, in 1.58×10^{-2} s on GPU. Batched inference at batch size 1024 reaches a per-sample cost of 1.7×10^{-5} s, equivalent to a throughput of $\sim 6.0 \times 10^{4}$ samples s^{-1}. The coupled HF solver requires 2266 s ($\sim$38 min) per case. The resulting speedups are $\approx 1.43 \times 10^{5}$ for one posterior-predictive draw, and $\approx 1.37 \times 10^{8}$ for batched per-sample inference. A 20,000-sample forward-UQ pass requires $\sim$0.3 s of GPU time excluding I/O, against 4.5×10^{7} s of sequential HF cost.

This speedup enables the full probabilistic workflow—20,000-sample Monte Carlo propagation, 50-replication Sobol analysis, and 18-case MCMC calibration—to run within minutes of wall-clock time, making iterative coupled-model exploration computationally tractable for HPR systems with significant material uncertainty (Fig. 6).

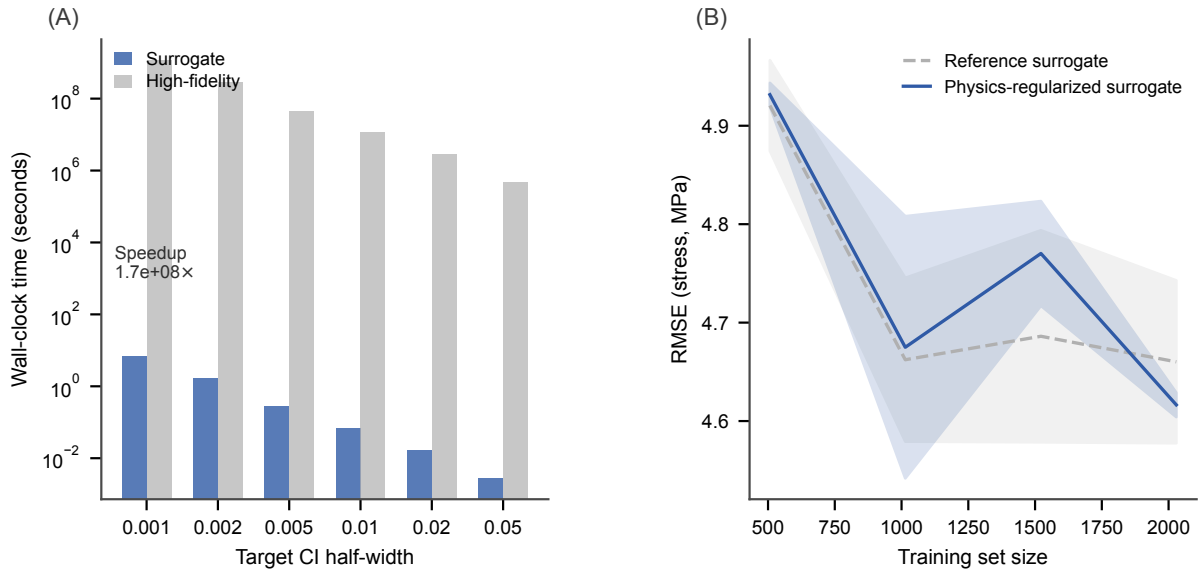

Figure 6: **Computational efficiency.** (**A**) Speedup of BNN surrogate relative to coupled HF solver: single posterior-predictive draw ($1.43 \times 10^{5}\times$) and batched per-sample inference ($1.37 \times 10^{8}\times$). (**B**) Data efficiency: per-output R^2 as a function of training fraction (Supplementary Note G).

3.5 Limitations

The surrogate is trained on data from a single HPR design point. All quantitative results describe this specific configuration only; a different geometry, fuel enrichment or operating envelope would require regenerating the training set and retraining the BNN. The methodology itself is not design-specific: the same workflow of coupled-data generation, BNN training and three-stage probabilistic analysis is directly applicable to other reactor types and other coupled multi-physics

systems.

The MCMC calibration fixes four parameters (E_{slope}, T_{ref}, k_{slope}, ν) at their true values and calibrates the remaining four. Extending to a full eight-parameter calibration would require additional independent observations to constrain the expanded parameter space. The reported posterior contraction is therefore conditional on this four-parameter scope and should not be directly extrapolated to the unconstrained eight-dimensional case.

The k_{eff} surrogate degrades on out-of-distribution feature-tail subsets: on the α_{base} tail, k_{eff} R^2 drops to $\approx$0.31 versus 0.849 in-distribution, while stress retains $R^2 \geq 0.93$ across all tested tails. The Sobol and posterior conclusions for k_{eff} therefore hold only within the training range of the dominant reactivity parameter (Supplementary Fig. S4).

All posterior summaries in Section 2.5 are computed on the surrogate side. An independent high-fidelity re-evaluation at the posterior-mean parameter vectors (Supplementary Note J; 18 cases $\times$ 3 posterior labels = 54 HF runs) confirms consistency: stress MAE at the posterior mean is 5.65 MPa (4.5% relative) and the [5%, 95%] HF envelope covers all 18 true stress values. The present study uses synthetic observations constructed from the high-fidelity solver because no real operational data are available for this design. If experimental or monitoring data become available in the future, the same posterior calibration workflow can ingest real observations directly.

4 Conclusion

This work demonstrates that a single BNN posterior predictive distribution can serve as the shared probabilistic layer on which forward propagation, sensitivity attribution and posterior calibration all operate—without retraining or recalibration between analyses. Applied to a coupled OpenMC–FEniCS heat-pipe reactor model, the framework yields three linked findings. First, forward Monte Carlo propagation shows that two-way coupling damps stress variability by $\approx$30% relative to a decoupled prediction—a reduction invisible to single-physics analyses. Second, Sobol decomposition reveals that stress and k_{eff} are governed by largely separate parameter groups ($E_{\text{intercept}}$ and α_{base}, respectively), so the two observables carry non-redundant information for inverse analysis. Third, MCMC calibration contracts the material-parameter distribution toward observation-compatible regions with 90%-CI coverage 0.861 across 18 held-out benchmarks, and the posterior-mean BNN stress tracks the true high-fidelity stress at MAE 10.4 MPa over the 102–198 MPa range.

The surrogate is roughly five orders of magnitude cheaper per evaluation than a coupled high-fidelity solve. This cost reduction makes the full uncertainty workflow—propagation, sensitivity and calibration—executable end-to-end on a single workstation. More broadly, the posterior-predictive framework converts analyses that were previously gated by repeated high-fidelity solves into routine computations on the same probabilistic object. The workflow is not specific to this reactor design: any coupled multi-physics system for which a BNN surrogate can be trained admits the same three-stage analysis pipeline.

5 Methods

5.1 Material uncertainty model

The coupled HPR simulation involves eight material properties of the SS316 monolith grouped by physical function: structural mechanics ($E_{\text{intercept}}$, E_{slope}, ν), thermal deformation (α_{base}, α_{slope}), and thermal transport (k_{ref}, T_{ref}, k_{slope}). All eight are assigned independent uniform priors with half-width 10% of the nominal design value. The same prior distribution is used throughout: as the sampling distribution in forward UQ, as the input domain for Sobol analysis, and as the prior in posterior calibration. Full parameter definitions, nominal values and constitutive-law relationships are in Supplementary Note E and Supplementary Table S1.

5.2 Coupled high-fidelity simulation

The high-fidelity workflow couples OpenMC Monte Carlo neutronics with the FEniCS finite-element solver through a Picard fixed-point iteration scheme[9]. Two fidelity levels are retained: a decoupled single-pass prediction before coupling convergence and a coupled steady-state prediction after full convergence. Each HF evaluation produces 15 scalar outputs. A dataset of $n = 3418$ samples was generated by Latin hypercube sampling; samples were split into training (2339), validation (501) and test (501) sets with a fixed random seed (seed = 2026). The BNN maps directly from the 8 input parameters to the 15 outputs without resolving intermediate coupling fields. Details of the iteration scheme, convergence criteria and output definitions are in Supplementary Note E.

5.3 Bayesian neural network surrogate

The surrogate is a BNN with mean-field variational inference (MFVI, Bayes-by-Backprop[10]). The network uses a fully connected backbone with two output heads: a mean head and a log-variance head that models heteroscedastic observation noise. Training minimises the negative evidence lower bound (ELBO). The posterior predictive distribution at a new input is approximated by Monte Carlo averaging over M weight samples from the trained variational posterior: at inference $M = 50$, for Sobol analysis $M = 30$, and for the MCMC likelihood $M = 20$. Hyperparameters were optimised via Optuna (Supplementary Table S2). Physics-prior monotonicity constraints are incorporated as an auxiliary loss term during training. Full ELBO derivation, KL annealing schedule and architecture details are in Supplementary Note A.

5.4 Forward uncertainty propagation

Forward UQ draws $N = 20{,}000$ Latin hypercube samples from the joint prior and evaluates the BNN posterior predictive mean at each sample. Using the predictive mean rather than a full predictive draw isolates parametric uncertainty from the BNN's own epistemic and aleatoric spread; the predictive-draw variant is reported in Supplementary Note D.

5.5 Sobol sensitivity analysis

Global sensitivity is assessed through variance-based Sobol indices estimated with the Saltelli cross-matrix scheme ($N_S = 4096$ per matrix, 40,960 surrogate evaluations per replication). The analysis is repeated 50 times; a factor is treated as a stable contributor only if the lower bound of its 90% CI exceeds zero. Methodological details including the Jansen estimator definition are in Supplementary Note B.

5.6 Posterior calibration

Posterior calibration uses Bayesian inference: $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})\, \pi(\boldsymbol{\theta})$, with the likelihood evaluated through the BNN posterior predictive mean. Four parameters are calibrated ($E_{\text{intercept}}$, α_{base}, α_{slope}, k_{ref}), selected on the basis of their Sobol ranking; the remaining four are fixed at their true values. The 18 benchmark cases are drawn from the test split (6 low, 6 near, 6 high stress). Observations are the true HF outputs of 5 primary coupled quantities perturbed by 2% multiplicative Gaussian noise. The posterior is sampled by Metropolis–Hastings MCMC with 4 independent chains (8000 iterations, burn-in 2000, thinning 5). Implementation details and convergence diagnostics are in Supplementary Note C.

Data availability

The coupled simulation dataset and fixed train/validation/test split used in this study are available at [repository URL upon acceptance].

Code availability

The BNN training, evaluation and analysis code are available at [repository URL upon acceptance].

References

1. McClure PR, Poston DI, Dixon DD, Cooley JW. Design of megawatt power level heat pipe reactors. LANL, LA-UR-15-28840; 2015.
2. Stauff NE, et al. High-fidelity multiphysics modeling of a heat pipe microreactor using BlueCrab. *Nucl. Sci. Eng.* 2024.
3. Miao Y, et al. Multiphysics simulation of KRUSTY warm critical experiments using MOOSE tools. *Nucl. Sci. Eng.* 2025.
4. Aldebie F, et al. Thermal-mechanical safety analysis of heat pipe micro reactor. *Nucl. Eng. Des.* 2024;420:113003.
5. Jeong MJ, et al. Multiphysics analysis of heat pipe cooled microreactor core. *Front. Energy Res.* 2023;11:1213000.
6. Slaughter AE, et al. MOOSE Stochastic Tools. *SoftwareX*. 2023;22:101345.
7. Dhulipala SLN, et al. MOOSE ProbML. *J. Comput. Sci.* 2026;94:102776.

8. Lim JY, et al. A hybrid surrogate modeling framework for the Digital Twin of a FHR. *Nucl. Eng. Des.* 2025;433:113690.

9. Zhang T, Chen Y, Wang S, Liu X. Design-oriented multiphysics analysis and surrogate modelling of a heat-pipe-cooled reactor. *Energy.* 2025. (in press)

10. Blundell C, et al. Weight uncertainty in neural networks. ICML 2015; pp. 1613–1622.

11. DOE Office of Nuclear Energy. What is a microreactor? 2021.

12. NASA. Fission Surface Power. 2024.

13. IEA. Electricity 2025: Analysis and forecast to 2027. Jan 2025.

Acknowledgements

The authors gratefully acknowledge the support provided by the National Natural Science Foundation of China [12175138]; Shanghai Rising-Star Program; and LingChuang Research Project of China National Nuclear Corporation.

Author contributions

Y.C. developed the BNN surrogate model, designed and performed the experiments, analysed the results and drafted the manuscript. S.W. contributed to the coupled simulation framework. X.L. and T.Z. conceived and supervised the project, provided scientific guidance and revised the manuscript. All authors discussed the results and approved the final version.

Competing interests

The authors declare no competing interests.